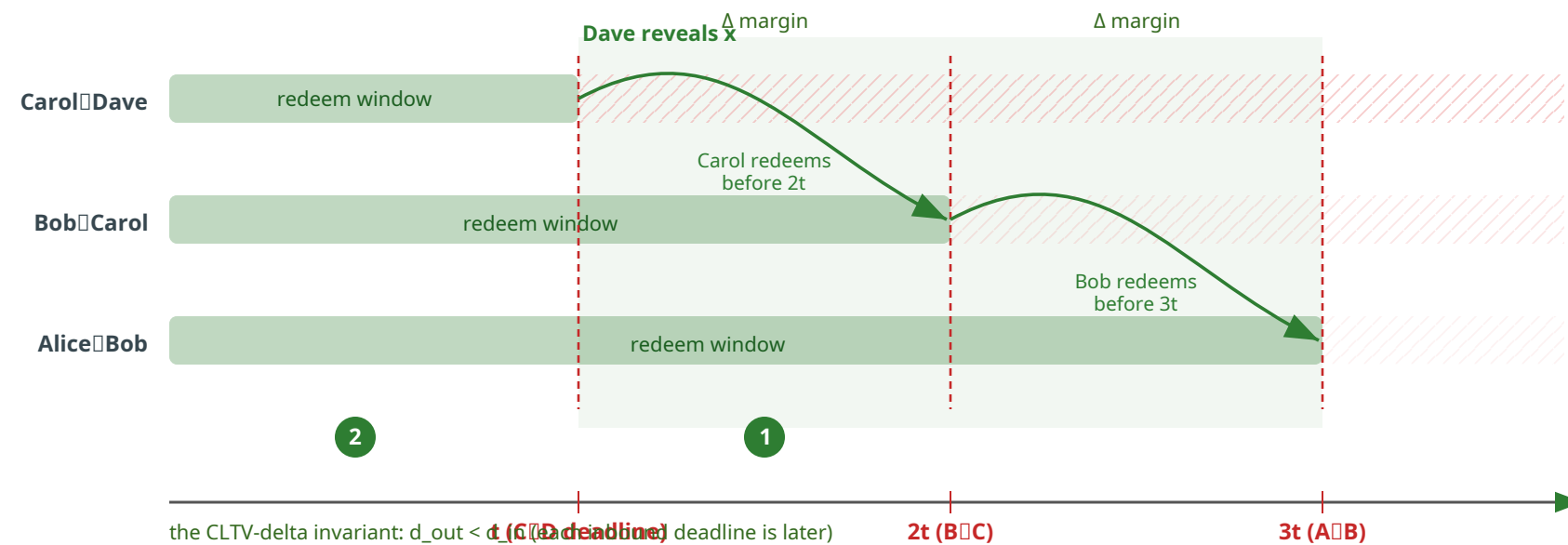
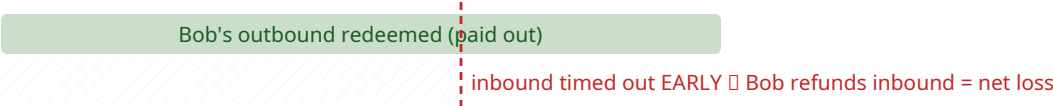


Staggered CLTV timelocks — timing safety (block height Δ)

covers `cltv_blocks.spthy` · `gaps.spthy` · `t2b_attack.spthy` — see rail Δ



3 WITHOUT the liveness assumption T2b (`t2b_attack.spthy`)
adversary fires the $A \sqcup B$ inbound timeout early $\square$ Bob's outbound is redeemed but his inbound refunds $\square$ Bob loses.



Two-sided certificate: race reachable here (T2b removed) $\square$ blocked in `multihop.spthy` (T2b_Counterexample_Blocked). So T2b is load-bearing.

Timing results live in separate abstractions; online claiming remains an explicit liveness assumption.

LEMMAS · timing theories

- 1 `cltv_blocks.spthy` — windows EXIST**
 $CLTV_Gap_Is_Positive \square \cdot Claim_Window_Nonempty \square$
 $Staggered_Path_Safe \square$ (derives $d_{out} < d_{in}$ from block num)
- 2 `gaps.spthy` — claim-in-time + lifecycle**
 $Claim_Window_Exists \square \cdot Outcome_Exclusive \square$
 $Transitive_Preimage_Before_Upstream_Deadline \square$
 $No_HTLC_After_Close \square \cdot Redeem_Reachable / Refund_Reachable \square$
- 3 `t2b_attack.spthy` — the race (T2b removed)**
 $Early_Timeout_Race \square$ (reachable without T2b)
pairs with `multihop`'s `T2b_Counterexample_Blocked` $\square$
 $\square$ together: proof that T2b is necessary, not assumed away

green = timing safety proved · red = the race, shown reachable to justify the assumption.
*`cltv_blocks` proves that the assumed deadline stagger leaves a claim w
`gaps` adds the block-clock lifecycle safety.*